IMPORTANT DIFFERENCES between LOG files from [mcr_2405] –vs– [2603] versions of DMON firmware/d3 programs:

- **[mcr_2405]** version has **41 characters** between "$DUTY" and "*" in each line that get decoded *([2603] has 49 characters)*
- **[mcr_2405]** version uses **MILLIseconds** represented by **4-character** field *([2603] uses microsec 8-char field; **1 milisec = 1000 microsec**)*

UVI2_AroundSTX_20250306 mission used **[mcr_2405]** version DMON update release
*(firmware **LF2_HF240d1_5m.bin**, d3host.exe **version 2.2**, d3read.exe **version 2.0**, and reformat_dmon_wav_files.sav **version 1**)*

Example $DUTY line in log file (from UVI2_AroundSTX_20250306_050.log):
1742680666,1594,$DUTY,67DF244701F967DF3257000520000018300 6DDD00*07

*The **41** characters between "$DUTY" and "*" in each line can be decoded as follows…*

Characters 1-8 (n=8): Start datetime in Unix epoch UTC
67DF2447 hexadecimal = **1742677063** decimal (sec Unix epoch) = **2025-03-22 20:57:43** start datetime UTC

Characters 9-12 (n=4): Milliseconds after start dt in Unix epoch UTC
01F9 hex = **505** decimal (millisec) →
2025-03-22 20:57:43 UTC + 505 millisec = **2025-03-22 20:57:43.505** precise start dt UTC

Characters 13-20 (n=8): End datetime in Unix epoch UTC
67DF3257 hex = **1742680663** decimal (sec Unix epoch) = **2025-03-22 21:57:43** end datetime UTC

Characters 21-24 (n=4): Milliseconds after end dt in Unix epoch UTC
0005 hex = **5** decimal (millisec) →
2025-03-22 21:57:43 UTC + 5 millisec = **2025-03-22 21:57:43.500** precise end dt UTC

Character 25 (n=1): Hydrophone numeric ID **2 = LF hydrophone, 3 = HF hydrophone, 1 = MF hydrophone*
2 numeric ID (does NOT require decoding) = LF hydrophone

Characters 26-33 (n=8): Sequential recording session number for **specific hydrophone (LF/HF)* during mission
00000183 hex = **387** decimal = **387th** recording session for LF hydrophone during mission

Characters 34-41 (n=8): Number of samples in current recording session
006DDD00 hex = **7200000** decimal = **7,200,000** total samples in the 387th LF recording session

OPTIONAL: Check if n samples matches expected value calculated from duration and sample rate
- Duration = 2025-03-22 20:57:43.505 UTC to 2025-03-22 21:57:43.500 UTC = 3,599.995 sec
- Sample rate = 2 kHz = 2,000 Hz = 2,000 samples/sec
- Expected total samples = 3,599.995 sec × 2,000 samples/sec = 7,199,990 samples
  OFF by 10.0 samples

EXTRA EXAMPLE FROM HF HYDROPHONE $DUTY LINE

Example $DUTY line in log file (from UVI2_AroundSTX_20250306_050.log):

1742679226,694,$DUTY,67DF2C7B03E867DF2CB703E430000121F00DBBA00*7E

*The **41** characters between "$DUTY" and "*" in each line can be decoded as follows…*

Characters 1-8 (n=8): Start datetime in Unix epoch UTC

67DF2C7B hexadecimal = **1742679163** decimal (sec Unix epoch) = **2025-03-22 21:32:43** start datetime UTC

Characters 9-12 (n=4): Milliseconds after start dt in Unix epoch UTC

03E8 hex = **1000** decimal (millisec) = 1 sec →

2025-03-22 21:32:43 UTC + 1000 millisec = **2025-03-22 21:32:44.000** precise start dt UTC

Characters 13-20 (n=8): End datetime in Unix epoch UTC

67DF2CB7 hex = **1742679223** decimal (sec Unix epoch) = **2025-03-22 21:33:43** end datetime UTC

Characters 21-24 (n=4): Milliseconds after end dt in Unix epoch UTC

03E4 hex = **996** decimal (millisec) →

2025-03-22 21:33:43 UTC + 996 millisec = **2025-03-22 21:33:43.996** precise end dt UTC

Character 25 (n=1): Hydrophone numeric ID **2 = LF hydrophone, 3 = HF hydrophone, 1 = MF hydrophone*

3 numeric ID (does NOT require decoding) = HF hydrophone

Characters 26-33 (n=8): Sequential recording session number for *specific hydrophone (LF/HF) during mission

0000121F hex = **4639** decimal = **4,639**th recording session for HF hydrophone during mission

Characters 34-41 (n=8): Number of samples in current recording session

00DBBA00 hex = **14400000** decimal = **14,400,000** total samples in the 4,639th HF recording session